Smart City Energy Optimization – Final Roles & Requirements

🟦 Member 1 — ADF + Data Lake Engineer

Role: Responsible for Raw → Bronze ingestion + API ingestion + Scheduling + Alerts

Day 1

- Create ADLS folders: Raw / Bronze / Silver / Gold
- Create Linked Services:
    - ADLS
    - Weather API
    - Traffic API

Day 2

- Pipeline 1: **Ingest_Energy_Batch**
    - Add parameters
    - Add metadata columns
    - Configure mapping

Day 3

- Pipeline 2: **Ingest_Weather_API**
- Pipeline 3: **Ingest_Traffic_API**
- Create Daily Triggers

Day 4

- Add Error Handling: Alerts + Retry
- ADF Monitor checks
- Export ADF JSON (deliverable)

🟩 **Member 2 — Databricks Silver Engineer**

Role: Bronze → Silver Cleaning + Joining + Feature Engineering

Day 1

- Create notebooks
- Mount ADLS
- Load Bronze tables
- Schema validation

Day 2

- Cleaning rules
    - Remove duplicates
    - Fix timestamps
    - Standardize strings
    - Fix missing values

Day 3

- Feature Engineering
    - peak_hour_flag
    - weekend_flag
    - rolling_avg_7d
    - weather_index
    - congestion_index

Day 4

- Join energy + weather + traffic
- Build **silver.energy_enriched**
- Export Silver tables
- Create documentation

🟧 Member 3 — ML + Gold Modeling Engineer

Role: Build Gold Layer + Forecasting ML + Optimization
Tools: Azure ML Designer + Synapse SQL + MLflow

⭐ Day 1 — ML Pipeline Preparation (Azure ML Designer)

- Create Data Asset → **silver_master_table**
- Connect to ADLS Gen2
- Add Data Preparation Blocks:
  - Import Data
  - Select Columns (timestamp, region_id, energy_consumption, temperature, humidity, traffic_congestion)
- Convert to Time-Series Dataset:
  - frequency = hourly
  - group by = region_id
  - target column = energy_consumption
  - time column = timestamp
- Add Missing Value Handling (Clean Missing Data — Linear Interpolation)

⭐ Day 2 — Train Forecasting Model

- Use Forecasting block
- Algorithms: ARIMA / Prophet / Auto-ARIMA / Exponential Smoothing
- Connect Blocks: Time-Series → Train Model → Score Model
- Evaluate using:
  - MAPE
  - RMSE
  - MAE

- Export forecast results → **gold.energy_forecast**

⭐ **Day 3 — Build Gold Tables**

Gold Table 1 — gold.energy_forecast

- timestamp
- region_id
- predicted_consumption
- actual_consumption (optional)
- prediction_error

Gold Table 2 — gold.risk_zones

Logic:

- If prediction > historical_peak ⇒ **High Risk**
- Near peak ⇒ **Medium Risk**
- Below ⇒ **Low Risk**

Gold Table 3 — gold.optimization_recommendations

- region_id
- predicted_peak_time
- recommendation
    - Shift non-critical loads
    - Increase solar storage usage
    - Reduce HVAC consumption
    - Balance across zones

⭐ **Day 4 — Optimization Engine**

- Peak Shifting (detect high demand hours)
- Critical Region Priority (e.g., hospitals)

- Load Balancing between zones
- Export: **gold.optimization_recommendations**

⭐ Day 5 — MLOps + MLflow + Synapse SQL Views

- Register model in MLflow
- Track metrics: MAPE / RMSE
- Track experiment runs
- Create SQL Views in Synapse:
  - view_energy_forecast
  - view_risk_zones
  - view_optimization_recommendations
- Make them available for Power BI

🟦 *Additional Notes for Member 3 → Impact on Member 4*

- MUST mention **Azure ML Designer** in documentation.
- Add a Presentation Slide:
  - "AI Pipeline — Azure ML Designer"
  - Screenshot of Designer canvas
  - Score model output
  - MLflow experiment screenshot

🟩 Member 4 — Power BI + Documentation & Presentation

Role: Dashboards + Architecture + Final PDF + Presentation Slides

Day 1 — Power BI Model

- Connect to Synapse Serverless
- Load:
  - gold.energy_forecast

  - gold.risk_zones
  - gold.optimization_recommendations

Day 2 — Visuals

- Region energy usage
- Weather impact on energy
- Traffic correlation

Day 3 — Advanced Visuals

- Forecast vs Actual
- Risk Zone Heatmap
- Optimization Recommendations Panel

Day 4 — Documentation

- Architecture Diagram
- Dataflow Diagram
- Medallion Architecture
- Tools Used
  - (Add Azure ML Designer)
- Screenshots
- Data Dictionary

Day 5 — Final Presentation

- Storytelling
- Animations
- Screenshot of ML Designer pipeline
- MLflow metrics screenshots
- Export full Deliverables PDF